

Factor Graph

Shirin Ahmadi

What is the Factor Graph?

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- A factor graph is a mathematical concept.
- It is a bipartite graph representing the factorization of a function.
- This means breaking down a function into a product of smaller functions that depend on subsets of the variables the original function depends on.

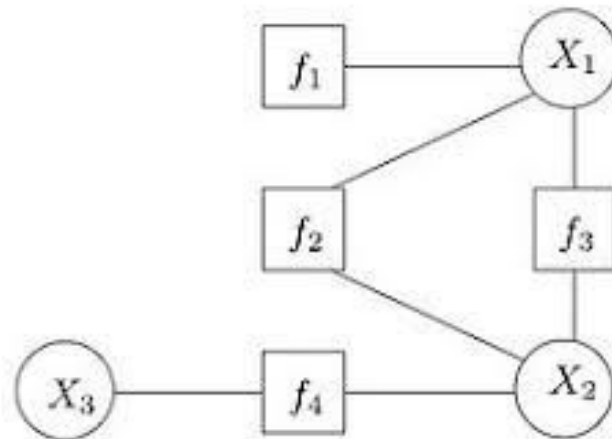
Factor Graph in Probability Distributions

- Factor graphs are valuable in representing joint probability distributions.
- They break down distributions into products of smaller distributions, each depending on fewer variables.
- Similar to Bayesian networks or Markov networks, factor graphs describe independence relations between variables and enable efficient inference algorithms.

Structure of Factor Graphs

Anatomy of the Factor Graph

- Factor graphs consist of nodes representing variables and factors representing individual functions.
- Factors connect nodes representing variables they depend on.
- The overall function of all variables can be expressed as a product of individual functions.



Representing Graphical Models as Factor Graphs

There are strong relationships between **Bayesian networks**, **Markov networks**, and vector graphs. We can show that these models can be effectively represented as a **Factor Graph**.



Implementation Details

Formal algorithms, such as **message passing** algorithms, are utilized within this framework.

Implementation

Message passing in Factor Graph

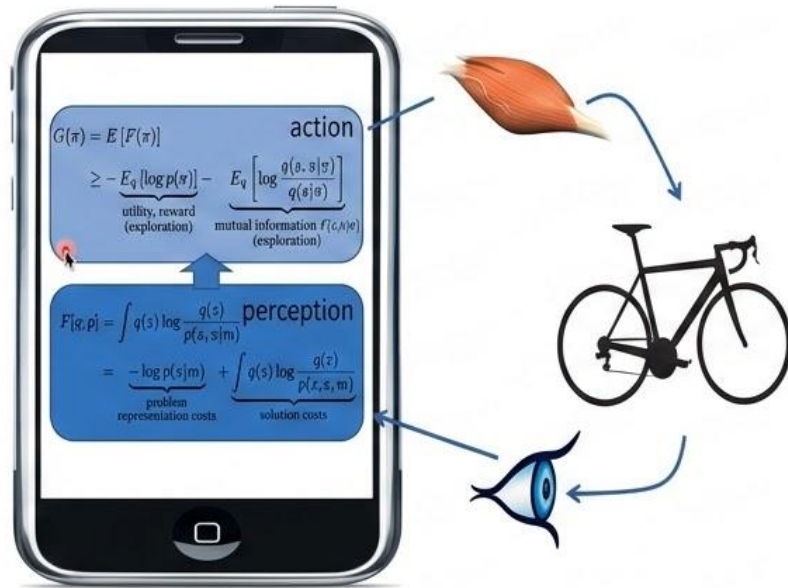
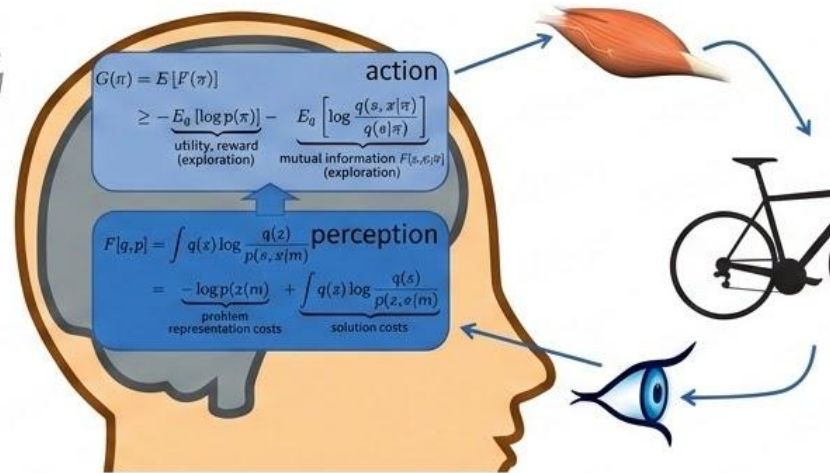
"Every good regulator of a system must be a model of that system."

Core Implication:

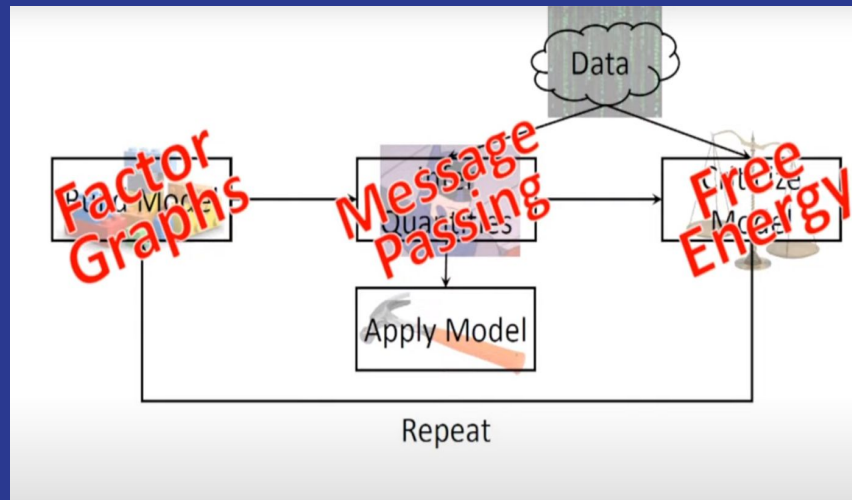
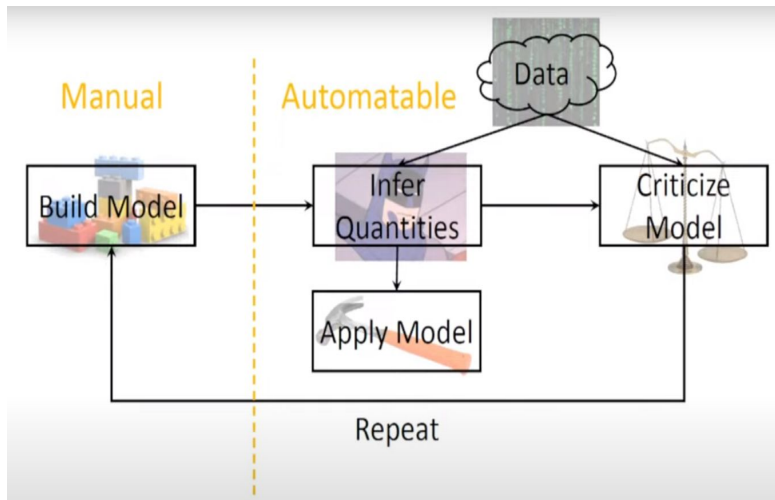
To build an agent that effectively regulates or manipulates an environment...

...we must first build a model of that system.

real-time



Model Design Cycle



Model Design Cycle

Model Proposal

Engineers conceptualize relationships between variables to propose a model representing system functions.

Inference

Statistical or probabilistic methods are used to estimate unknown values based on available observed data.

Critique

Critical evaluation of performance, focusing on predictive accuracy, consistency, and computational efficiency.

Reiteration/Rebuild

Revising model structure or adjusting parameters based on critique to improve overall performance.

Application

Satisfactory models are deployed in real-world scenarios to inform decisions and solve problems.

Message Passing & FFG

Message Passing

A computational method used in probabilistic models where information is exchanged between different parts of a system to update beliefs or make decisions.

Forney-style Factor Graph (FFG)

A specific type of factor graph well-suited for representing dynamic models and automating active inference through message passing.

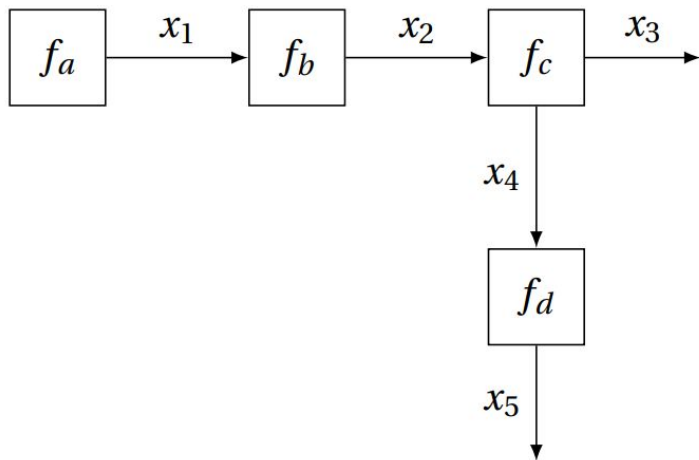


Automating Active Inference

FFGs provide the structural framework for efficient, localized computations through message exchanges.

Graph Design

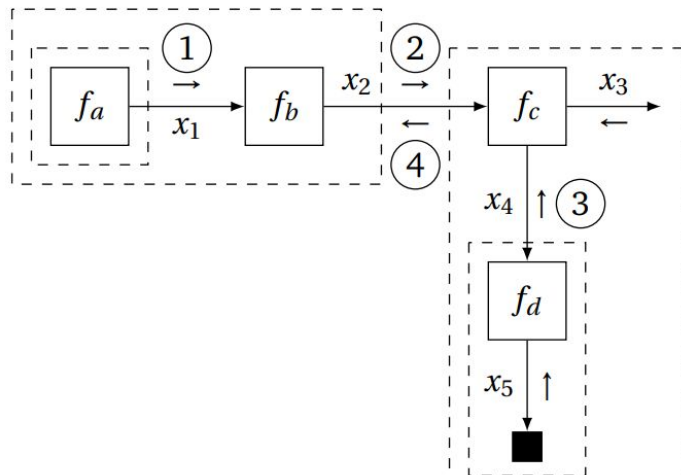
Forney-Style factor Graphs



- Edge correspond to variable and nodes correspond to the factors
- Description of the generative model with five variables and four factors

$$f(x_1, \dots, x_5) = f_a(x_1) f_b(x_1, x_2) f_c(x_2, x_3, x_4) f_d(x_4, x_5),$$

Inference in Factor Graphs



marginalization: integrating out variables except x_2 while considering the observed data x_5 .

- Challenge of integration in large models and the need for smart approaches.
- Introduction to message passing as a solution for automated inference.
- Nested solution approach: using previous messages in the computation of subsequent messages.
- Visualization of how message passing leads to the calculation of posterior beliefs over variables.

$$f(x_2 | x_5 = \hat{x}_5) \propto \int \cdots \int f(x_1, \dots, x_5) \cdot \delta(x_5 - \hat{x}_5) dx_1 dx_3 dx_4 dx_5 \quad (2.2a)$$

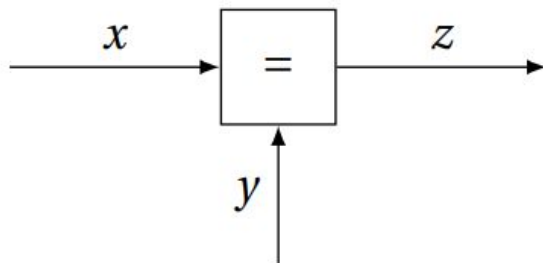
$$= \underbrace{\int f_a(x_1) f_b(x_1, x_2) dx_1}_{(1)} \underbrace{\iint f_c(x_2, x_3, x_4) \left(\int f_d(x_4, x_5) \cdot \delta(x_5 - \hat{x}_5) dx_5 \right) dx_3 dx_4}_{(4)}$$

Summarize all the information
inside this box (3)

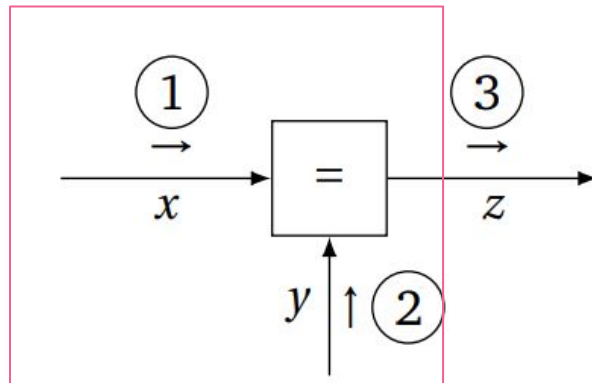
For the expression $(\delta(x_5 - \hat{x}_5))$, it acts as a

When the expression matches the observed value ($\hat{x}_5$), the term in the model becomes non-zero.

Equality constraint node



you can kind of recognize a proportional Bayesian rule in here



Compute message 3 by multiplying the two messages that are incoming from the left and from the bottom.

3- posterior

1- prior

2- likelihood

$$f_{=}(x, y, z) = \delta(z - x)\delta(z - y)$$

$$\begin{aligned}\mu_3(z) &= \iint \mu_1(x) \mu_2(y) f_{=}(x, y, z) \, dx \, dy \\ &= \mu_1(z) \mu_2(z) .\end{aligned}$$

Forney-Style Factor Graphs (FFGs)

An **FFG** is a type of undirected graph, denoted as $G(V, E)$, where:

V represents the set of **nodes**.

E represents the set of **edges** connecting the nodes.

Neighboring Relationships

The **neighboring edges** of a node $a \in V$ are denoted as $E(a)$.

For an edge $i \in E$, the notation $V(i)$ refers to all neighboring nodes connected by that edge.

Forney-Style Factor Graphs: Subgraphs

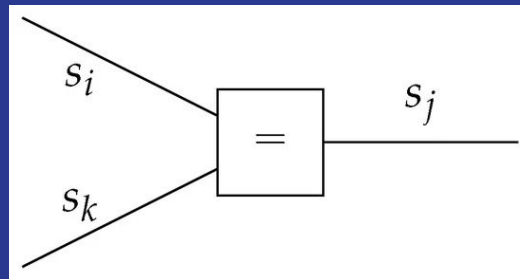
Subgraph Types

- **Edge-induced** $G(i)=(V(i),i)$:
Includes all nodes connected to a specific edge i .
- **Node-induced** $G(a)=(a,E(a))$:
Includes a node a and all its connected edges.
- **Local Subgraph** $G(a,i)=(V(i),E(a))$:
Collects nodes and edges around a specific edge i and node a .

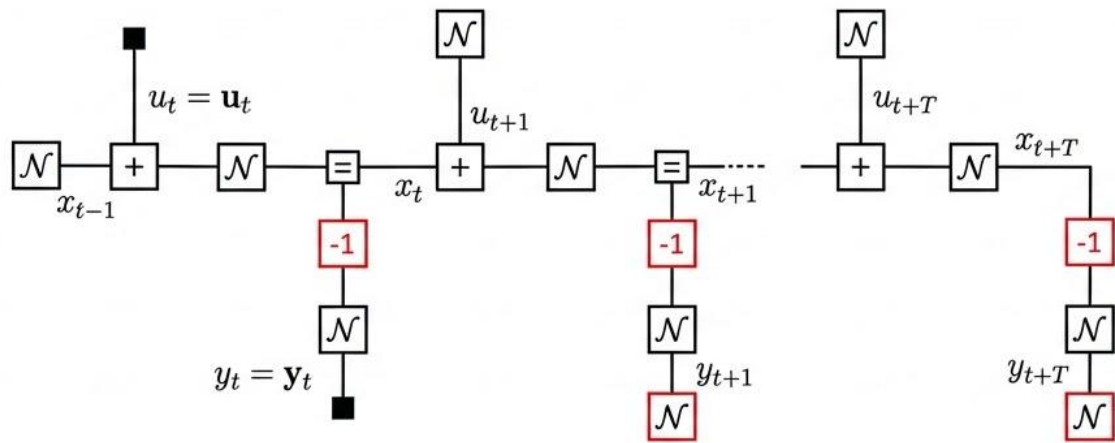
- An FFG can represent a **factorized function** $f(s)$, which is expressed as a product of factors:

$$f(s) = \prod_{a \in V} f_a(s_a)$$

- Here, s_a represents the set of variables that are arguments of the factor f_a .
- In an FFG, a node $a \in V$ corresponds to a factor f_a , and the neighboring edges $E(a)$ correspond to the variables s_a .

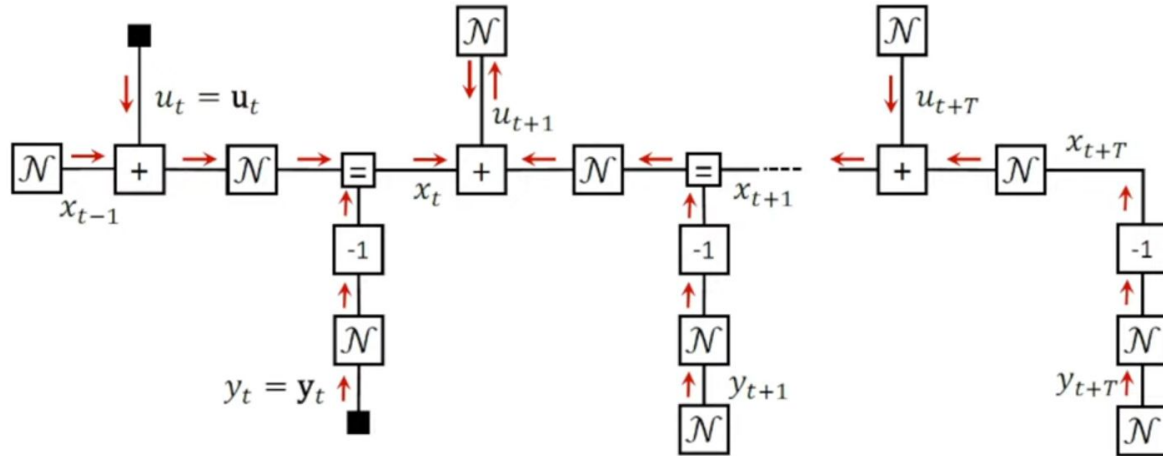


Bayesian Thermostat



Inference for Controls

$$F[q_t] = \left\langle \log \frac{q_t(y_{>t}, x, u_{>t})}{f(y, x, u)} \right\rangle_{q_t}$$



van de Laar & de Vries (2019). Simulating active inference processes by message passing. *Front. in Robotics and AI*, 6, 20.

Reference Materials

Key Publications & Resources

- **Active inference for Robot control: A Factor Graph Approach**
- **A Factor Graph Description of Deep Temporal Active Inference**
- Message passing-based inference in an autoregressive active inference agent
- ActInf ModelStream [#004.1](#) ~ "Implementing Active Inference by Message Passing in a Factor Graph"

Full Article Access:

<https://www.frontiersin.org/journals/robotics-and-ai/articles/10.3389/frobt.2019.00020/full>